

1 More convergence tests

Question 1. Determine whether the following series converge or diverge. State which test you used.

1. $\sum_{n=1}^{\infty} (-1)^n \cdot 2^{1/n}$ Divergence test $\lim_{n \rightarrow \infty} (-1)^n \cdot 2^{1/n}$ DNG
2. $\sum_{n=1}^{\infty} \frac{5 \cdot 2^{n+1}}{3^{n-1}}$ Geometric series $\frac{5 \cdot 2^{n+1}}{3^{n-1}} = 20 \left(\frac{2}{3}\right)^{n-1}$, converges between $(-1, 1)$
3. $\sum_{n=1}^{\infty} \cos(n) 2^{-(n^2)}$ Converges absolutely by comparing with 2^{-n} . $|\cos(n) 2^{-(n^2)}| \leq 2^{-n}$
4. $\sum_{n=1}^{\infty} \frac{\cos(\pi n)}{\sqrt{n}}$ Converges. $\frac{\cos(\pi n)}{\sqrt{n}} = \frac{(-1)^{n+1}}{\sqrt{n}}$ alternating series
5. $\sum_{n=1}^{\infty} \frac{n!}{\sqrt{(2n)!}}$ Converges, ratio test. $\frac{(n+1)!}{\sqrt{(2n+2)!}} \cdot \frac{\sqrt{(2n)!}}{n!} = \frac{\sqrt{(n+1)!}}{\sqrt{(2n+2)!}}$
 $\rightarrow \frac{1}{2} < 1$
6. $\sum_{n=1}^{\infty} \left(\frac{5}{4}\right)^n \left(\frac{4n}{5n+1}\right)$ Divergence test. $\lim_{n \rightarrow \infty} \left(\frac{5}{4}\right)^n \left(\frac{4n}{5n+1}\right) = \infty$
7. $\sum_{n=1}^{\infty} \sin^2\left(\frac{1}{n}\right)$ Converges, compare with $\frac{1}{n^2}$. $\lim_{n \rightarrow \infty} n^2 \sin^2\left(\frac{1}{n}\right) = 1$
 $= \lim_{n \rightarrow \infty} (n \sin\left(\frac{1}{n}\right))^2 = 1$
8. $\sum_{n=1}^{\infty} \sin^n\left(1 + \frac{1}{n}\right)$ Converges, root test. $\lim_{n \rightarrow \infty} \sin\left(1 + \frac{1}{n}\right) = \sin(1) < 1$
9. $\sum_{n=1}^{\infty} \left(1 - \frac{1}{n}\right)^{(n^2)}$ Converges, root test. Note $x^{(n^2)} = (x^{n \cdot n}) = (x^n)^n$
 $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = e^{-1} < 1$
10. $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ Converges, comparing with $\frac{2}{n^2}$. Alternatively by ratio test
 $\frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{n^n}{(n+1)^n} = \left(\frac{n}{n+1}\right)^n \rightarrow e^{-1} < 1$

Question 2. The following boxed argument falsely concludes that the series $\sum_{n=1}^{\infty} \frac{1+2(-1)^n}{n}$ converges. Identify the mistake(s) and provide a correct argument for the divergence of the series.

Notice that

$$\sum_{n=1}^{\infty} \frac{1+2(-1)^n}{n} = -\frac{1}{1} + \frac{3}{2} - \frac{1}{3} + \frac{3}{4} - \frac{1}{5} + \frac{3}{6} - \dots$$

To apply the AST, the absolute value of the series must be decreasing. $(1, \frac{3}{2}, \frac{1}{3}, \frac{3}{4}, \frac{1}{5}, \frac{3}{6}, \dots)$ is not.

is an alternating series and the limit of $-\frac{1}{n} \leq \frac{1+2(-1)^n}{n} \leq \frac{3}{n}$ as $n \rightarrow \infty$ is 0 by the sandwich theorem. Thus, it converges by the alternating series test.

convergent series + divergent series = divergent series
 $\sum \frac{2(-1)^n}{n}$ is convergent, $\sum \frac{1}{n}$ is divergent

Question 3 (More practice). Determine whether the following series converge or diverge. State which test you used.

1. $\sum_{n=2}^{\infty} \frac{1}{(\ln n)^3}$ Diverges, compare with $\frac{1}{n}$.
 $\lim_{n \rightarrow \infty} \frac{1}{(\ln n)^3} \cdot \lim_{n \rightarrow \infty} \frac{1}{n} = \lim_{n \rightarrow \infty} \frac{1}{n^3} = 0$
2. $\sum_{n=1}^{\infty} \frac{2^{(n^2)}}{n!}$ Divergent by divergence test.
 $\frac{2^{(n^2)}}{n!} = \frac{(2^n)^n}{n!} = \frac{2^n \cdot 2^n \cdots 2^n \cdot 2^n}{1 \cdot 2 \cdot 3 \cdots (n-1) \cdot n} \geq 2^n$
3. $\sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{\sqrt{n}}\right)$ Diverges, compare with $\frac{1}{\sqrt{n}}$.
 $\lim_{n \rightarrow \infty} \sqrt{n} \ln\left(1 + \frac{1}{\sqrt{n}}\right) = \lim_{x \rightarrow 0} x \ln\left(1 + \frac{1}{x}\right) = 1$
4. $\sum_{n=1}^{\infty} \frac{(\ln n)^n}{n}$ Divergent by divergence test.
 $\lim_{n \rightarrow \infty} \frac{(\ln n)^n}{n} = \lim_{n \rightarrow \infty} \frac{\ln(\ln n)^n}{\ln n} = \lim_{n \rightarrow \infty} \frac{n \ln(\ln n)}{\ln n} = \lim_{n \rightarrow \infty} \frac{\ln(\ln n)}{\ln n} = 0$
5. $\sum_{n=1}^{\infty} a_n$ where $a_n = \begin{cases} \frac{1}{n} & \text{if } n \text{ is a square number} \\ -\frac{1}{n^2} & \text{otherwise} \end{cases}$
 Converges, let $b_n = \begin{cases} \frac{1}{n} & \text{if } n \text{ is a square} \\ 0 & \text{o.w.} \end{cases}$
 $c_n = \begin{cases} 0 & \text{if } n \text{ is a square} \\ \frac{1}{n^2} & \text{o.w.} \end{cases}$
 $\sum c_n$ converges because $\sum c_n \leq \sum \frac{1}{n^2}$.
 $\sum b_n$ converges because $\sum b_n = \frac{1}{1} + 0 + \frac{1}{4} + 0 + \frac{1}{9} + 0 + \frac{1}{16} + 0 + \frac{1}{25} + 0 + \frac{1}{36} + \dots$
 $= \sum \frac{1}{n^2}$.
 Thus, $\sum a_n = \sum b_n - \sum c_n = \sum b_n - \sum c_n$ converges

$\ln\left(1 + \frac{1}{x}\right) \approx x$
 when x is small